Consistent but Blind?

Image removal does not measure image reliance in medical vision-language models

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Paper, code, data and every number

Background

Medical VQA benchmarks measure whether an answer is correct, not whether the model used the image. An answer derived from language priors can be correct on typical cases and fail precisely on atypical images. Consistency under rephrasing does not resolve this: a model that ignores the image can be *more* consistent, since no visual signal perturbs it.

The *consistent-but-dangerous* taxonomy (Sadanandan & Behzadan, CVPR-W 2026) therefore combines two per-answer criteria: stability under paraphrase, and whether the answer changes when the image is removed.

	changes without the image	**unchanged** without the image
stable under paraphrase	IDEAL	DANGEROUS
unstable under paraphrase	FRAGILE	WORST

A DANGEROUS answer appears reliable but does not depend on the image.

Question. Does the taxonomy's reliance criterion, **image removal**, rank models in the same order as **in-format probes**, which keep an image in the input?

Method

Models. Three pairs, each a medical model and the general-purpose model it was derived from (same architecture and size): **MedGemma-4B** and **MedGemma-1.5-4B** vs. **Gemma-3-4B**; **MedVLThinker-3B** vs. **Qwen2.5-VL-3B**.

Data. 3,365 questions on 907 images from SLAKE (CT, MRI, X-ray), VQA-RAD (radiology) and PathVQA (pathology); the primary set comprises 1,206 yes/no test questions.

Probes. Each probe alters only the image or the instruction. A model that uses the image should change its answer when the image changes; for each probe we report the *answer-change rate*.

Image removed	image tokens omitted; the taxonomy's criterion
Different image	a random image of the same dataset and modality
Black image	format preserved, content removed
Opposite-answer image	an image for which the *same question* has the opposite answer; the most direct test
Abstention permitted	image removed; the model may answer "unknown"

Scoring. P(yes) from the yes/no token logits in a single forward pass; the total yes/no probability identifies refusals ("please provide the image"). Identical prompts, bf16, greedy decoding; about 120,000 model calls on one 16 GB GPU; 95% CIs from a bootstrap over images.

Pre-registration. Endpoints were fixed before the main run, and each subsequent probe and model pair was registered (as a dated git tag) before it was run. Unsupported predictions are reported.

Finding 1: no in-format probe corroborates image removal



Reading the figure. Each point is the medical model minus its general-purpose base: the difference in answer-change rate under one probe. Values above zero indicate that the medical model responds more to the image.

Result. Image removal (top row) separates the two models in five of nine model-pair × dataset cells. In four of these, no in-format probe detects a difference; in the fifth (MedGemma-4B on SLAKE), every in-format probe shows a difference in the opposite direction.

5 of 9
cells in which image removal separates the models

0 of 5
of these corroborated by any in-format probe

Sources of the image-removal differences

Refusal. Without the image, **Gemma-3** declines on **29%** of questions, **MedGemma-4B** on 18%, **MedGemma-1.5** on 0.7%, and the Qwen-based models on none. Forced answers on declined questions are unrelated to the answers given with the image ($\kappa \approx 0$), so about half are classified as image-reliant: a higher refusal rate inflates apparent reliance.

Right: questions declined without the image. Forced yes/no format (top): Gemma-3 declines far more often. Abstention permitted (bottom): both models decline on ≥93%.

Agreement between prior and image. When a model's prior matches the image content, its answer does not change without the image, and image removal classifies it as image-independent. This accounts for the differences in the Qwen pair, in which neither model declines, and for MedGemma-4B's additional DANGEROUS answers, which are **correct**: +14.1 points DANGEROUS-and-correct, –0.4 DANGEROUS-and-incorrect.

Finding 2: the in-format advantage is confined to training data

Both MedGemma models were trained on SLAKE, where the in-format probes favoured them. We registered a test on **PathVQA**, which none of the medical models was trained on, before running it:

Follows the opposite-answer image medical – general, points	**SLAKE** training data	**PathVQA** held out
MedGemma-4B – **Gemma-3**	+48.0	+0.8 [–2.7, 4.2]
MedGemma-1.5 – **Gemma-3**	+36.5	–5.6 [–9.1, –2.0]
MedVLThinker – **Qwen2.5-VL**	–1.6	–0.9 [–3.0, 1.3]

Both MedGemma predictions were unsupported, so under the decision rule fixed in advance the hypothesis that they rely on the image more was **withdrawn**: the SLAKE difference reflects training exposure.

Recommendations for reliance audits

1. Do not compare models by a DANGEROUS share obtained by image removal; it reflects refusal and priors.
2. Assess reliance with in-format probes, preferably an image that requires the opposite answer.
3. Evaluate those probes on data outside each model's training set.
4. Report the chance floor, decompose by correctness, and permit abstention.

Corrections to the DANGEROUS share

- **Chance floor:** 36–42% of yes/no answers appear image-independent by chance alone.
- **Correctness:** the additional DANGEROUS answers are predominantly correct.
- **Error detection:** image reliance identifies errors at chance level (AUROC **0.51**; model confidence 0.71).

Limitations

Three model pairs establish that the disagreement occurs, not how often. The mechanistic accounts are post hoc; the held-out test covers pathology only. Paraphrase equivalence was judged by an LLM (a 50-item check by the first author, with AI assistance, accepted 88%). No clinical claims are made.

Sadanandan & Behzadan, CVPR-W 2026; Zafar et al. 2026; Lotfinia et al. 2026; Sellergren et al. (MedGemma) 2025.